A fourteen-positive-variable Diophantine consistency equation of degree 30

Research contribution for ProveIt

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1 One free scale and an index absorber: fourteen positive variables

A useful new tradeoff is a polynomial of exact degree 30 in fourteen strictly positive integer unknowns, with 87 binary arithmetic operations. The comparison is the existing fourteen-positive degree-52, 79-operation construction: degree decreases, cost increases. It also saves one variable relative to the incoming integral-root degree-24/90, 26/88 and 30/79 constructions. No global minimality or domination is asserted.

The idea is to free *only* the ordinate scale Y using a monic-root carrier, while retaining $X = wq$. The recovered divisibility $q^3 \mid Y$ then permits the adaptive $U = 4qY$ index absorber to delete its quotient witness. Attempting to merge both scale offsets into a single square root does not imply their separate divisibilities, and is not used here.

1.1 Literal polynomial

Use the entire unchanged compact tuple from the Integral Root packet: $\beta = 2^{590069}-1$, $K_0 = K$, M_C, and $M_F = M_{F,\mathrm{source}}$, with the latter already shifted. Its four actual integer payloads and binding are included. The fourteen positive unknowns are

$$J, F, \alpha, z, f, i, T, w, Y, r, \eta, \zeta, u, \gamma.$$

All remaining symbols are polynomial abbreviations:

$$\begin{aligned}
q &= \beta J + 1, & C &= q - F - \alpha,\\
X &= wq, & E &= XY,\\
k &= \eta + \zeta, & c &= kY + \eta,\\
a &= E + Y, & H &= 4a + 3,\\
\Delta &= a^2 + H, & \ell &= EkY,\\
A &= \ell(\ell + k) + 1, & G &= \Delta c^2 + 1,\\
U &= 4qY, & v &= k - R - 1,\\
Q &= \Delta(f^2 - 1), & I &= ic^2.
\end{aligned}$$

$$\begin{aligned}
R &= ((q-1)(q-F) + \alpha)(q^2-1) + (M_C + qM_F)J,\\
D_* &= U(X + ac) + H(U\gamma - v),\\
V &= f^2(cT - R) - c,\\
N_t &= (K_0 + w)C + q - F - z(q-1).
\end{aligned}$$

Define

$$\begin{aligned}
A_0 &= [q^3(\ell + r) + Y]^2 - q^6 A, \\
A_1 &= D_*^2 - U^2 G, \\
A_s &= I^2 - Q, \\
A_a &= (V + u)^2 - Qu(2V + u) - 1, \\
A_t &= N_t - 1.
\end{aligned}$$

The promised equation is

$$\boxed{P_{14} = A_0^2 + A_1^2 + A_s^2 + A_a^2 + A_t^2 = 0.}$$

There are no divisions, variable exponents or hidden existential variables in this literal expression. The five residuals can alternatively be used as a system of degrees $12, 14, 10, 15, 2$, with a shared 78-operation circuit.

1.2 Correctness of the new composition

At a zero, the rational square root $\tau = \ell + r + Y/q^3$ of the integer A is integral. Hence $q^3 \mid Y$; write $Y = sq^3$ with $s > 0$. This recovery precedes every application of an old scale-dependent bound. The transport equation gives the inherited whole-tuple bounds

$$q \geq 16, \quad C > 0, \quad 3q < R < q^4 - 2, \quad E \geq q^4 > R + 2.$$

The first Pell equation supplies $\tau = \chi_{P_0}(n)$ and $k = 2\psi_{P_0}(n)$, where $P_0 = 2XY^2 + 1$ and $n > 0$. The second residual makes $g = D_*/U$ integral. It is positive, because

$$D_* > U(X + ac) - Hk > ak(UY - 5) > 0.$$

Thus $g = \chi_{a+2}(p)$ and $c = \psi_{a+2}(p)$ for $p > 0$. The absorber gives $U \mid H(k - R - 1)$ without cancelling H. The adaptive modulus satisfies $U \geq 4q^4$, $U \mid 4XY^2$ and $\gcd(U, H) \leq 3$. For the last claim, $\gcd(Y, H) \mid 3$; every prime of q divides Y; and if $3 \mid q$ then $27 \mid Y$, so $H \equiv 3$ modulo 27. These are precisely the properties needed by the retained absorber proof. Moreover, $k \equiv 2n \pmod{4XY^2}$, with $U \mid 4XY^2$. Consequently

$$\mu = \frac{U}{\gcd(U, H)} > R + 1, \qquad 2n \equiv R + 1 \pmod{\mu}, \qquad n \geq (R + 1)/2 \geq 25.$$

The standard first/main Pell comparison gives $p > n$, and therefore $c > (a + 2)\Delta^2$, $c > 2p$ and $c > 2R$. These bounds justify the inherited *ordinary* strong-rank lemma for $(ic^2)^2 = \Delta(f^2 - 1)$, yielding a main Pell index m with $p \mid m$, $c \mid m$, $c^2 \mid f^2 - 1$ and $m \geq c > 2p$.

Set $y = V + u$. The auxiliary identity is exactly

$$QV^2 - (Q - 1)y^2 = 1.$$

Although V need not be positive, y is: the identity gives $|y| \geq |V|$, while $u = y - V > 0$ rules out $y < 0$; zero is impossible. With $T_{\text{old}} = fT$, the signed expression $V = c(T_{\text{old}} f - 1) - Rf^2$ is the inherited auxiliary expression. Its index proof gives $p = R$; the absorber then gives $n < R$ and $2n = R + 1$. Hence

$$h = \frac{k - R - 1}{E} > 0, \qquad \gamma_{\text{old}} = \gamma - \frac{k - R - 1}{U}$$

are integral and recover the original equations. The positivity of the latter is not inferred from integrality: $k \geq 4P_0 > X$, $c > k > X$ and $g > (a + 1)c$ imply $g - (X + ac) > c - X > 0$, so $\gamma_{\text{old}} > 0$. The inherited ordinary compiler therefore produces a finite ZFC refutation.

Conversely, a finite refutation supplies canonical compact-tuple witnesses. Convert the normalized strong ordinate by $i = \Delta i_0$, and use the absorber shift $\gamma = \gamma_{\mathrm{old}} + (k - R - 1)/U > 0$. The Integral Root positive-gap lemma gives $\tau > \ell + s$, so $r = \tau - \ell - s > 0$. The retained squared-modulus completeness proof chooses T_{old} divisible by f and $y > V$, giving positive $T = T_{\mathrm{old}}/f$ and $u = y - V$. This is an existential choice of canonical witnesses, not a same-coordinate identity for arbitrary old witnesses. All five residuals vanish. The imported compiler calibration and this composition prove in ZFC that P_{14} has a positive zero exactly when there is a syntactic ZFC refutation; negation yields the desired equivalence to Con(ZFC).

1.3 Degree and replay boundaries

Writing $Z = UX + H(U\gamma - v)$ gives exact identities

$$A_0 = q^6[r^2 + \ell(2r - k) - 1] + 2q^3(\ell + r)Y + Y^2,$$
$$A_1 = 2UacZ + Z^2 - U^2(Hc^2 + 1).$$

Thus their degrees are 12 and 14, not the naive uncancelled bounds. For the second, Z has leading term H_0R_0 of degree 7, where $R_0 = (\beta J)^3(\beta J - F)$; its term $H_0U_0\gamma$ has only degree 6. The auxiliary residual has degree 15, with leading form $2E_0^2R_0f^4u$, where $E_0 = w\beta JY$. Hence the unique degree-30 homogeneous part of P_{14} is

$$4w^4(\beta J)^{10}Y^4(\beta J - F)^2f^8u^2,$$

which is nonzero for the actual positive β. The primary circuit uses 47 multiplications, 24 additions and 16 subtractions; the cancellation certificate is a separate circuit, not the advertised evaluator.

The attached standard-library replay authenticates forty source pins and all four fixed integer payloads, verifies three universal sparse polynomial identities, checks liveness and costs, and compares independently coded literal formulas. Exact-degree specializations use the actual coefficients modulo three primes. Three isolated corruption controls are rejected, including under Python optimization. These are arithmetic and integrity audits, not a fresh proof-assistant endpoint; the infinite Pell and compiler inputs remain explicitly inherited mathematical proofs.